Multi-Tones Expert Challenge

The Tone library that is included with Arduino lets us control a passive buzzer (piezo) to produce different pitches. However, there is a limitation that **only one buzzer can make a tone at a time.** Even if you wire in multiple buzzers, telling a second buzzer to make a tone will stop any noise being made by the first buzzer.

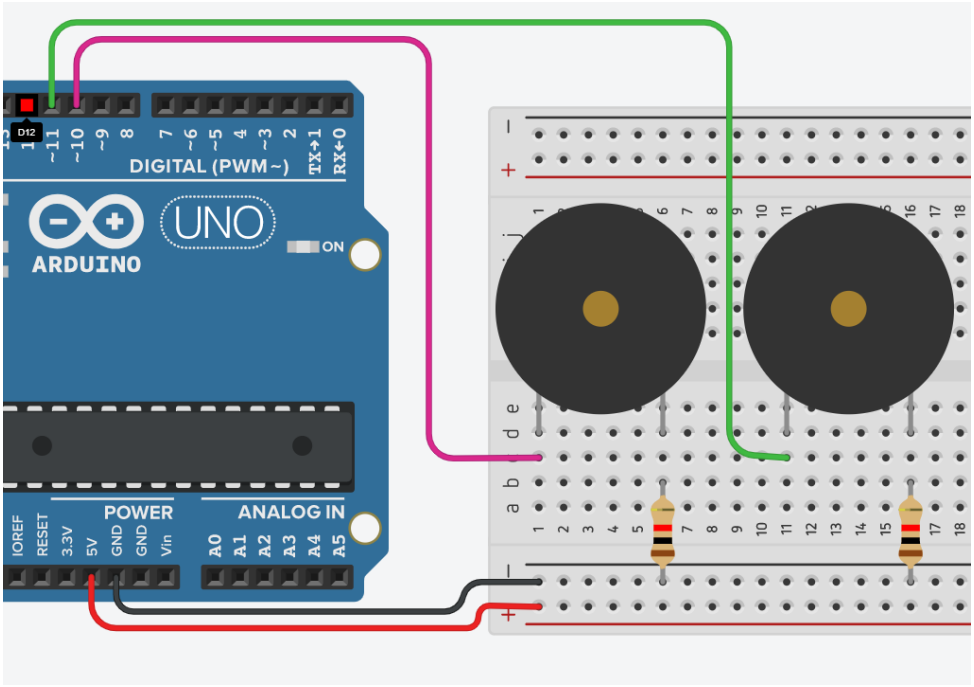
The reason for this gets into the nitty-gritty of how the Arduino microcontroller is making a tone in the first place – there is something called a **timer** that is used to produce a square wave, which is the sound we actually hear. There are a limited number of timers available on microcontrollers. On the **Arduino Uno** there are three, but one is reserved for the CPU, and one is used for tracking time with the millis() function, so normally there is only one free.

If, however, we were to sacrifice the millis() timer, it would be possible to generate two separate square waves simultaneously and make two concurrent sounds. As it turns out, a library has been created to allow just this!

Using an External Library

An external library is some code that we can make use of, but isn't included the Core Arduino Language. For this exercise, you'll download and install an alternative **tone library** that opens up the second timer (if you were using a different microcontroller, there may be some additional timers as well) so that we can make two tones.

Start by constructing the following physical circuit. There isn't any reason to use the simulator as we can't add external libraries of our own. At the start it should just be 2 piezo buzzers wired with a 1kΩ resistor for each, as shown below:



Next, look at the following GitHub page. This is where the external library is hosted, as well as the documentation about how to use it.

https://github.com/bhagman/Tone

To actually install the library, find the link to the latest version. This will let you download a ZIP file with the library in it. Then, from Arduino choose **Sketch → Include Library → add .ZIP Library**
Choose the file you just downloaded, and then you'll be able to use the library.

Challenge One – Play Two Constant Tones

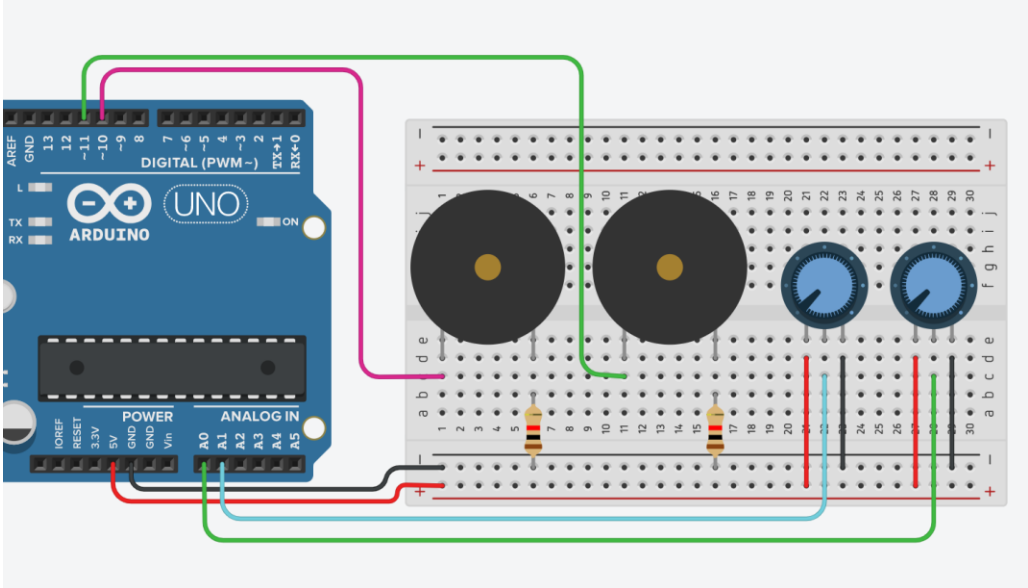
There is some sample code on the GitHub page that should give you an idea how to use this library. The essential step for telling Arduino that you want to make use of this library is the **include statement**, which appears at the top of the code:

```
#include <Tone.h>
```

Pick any two notes to play, and just have the Arduino play them together. This code for this should be quite short, as calling the **.play()** function once for each buzzer is all you need to do!

Challenge Two – Tone Mixing

Expand your circuit and code so that you have two potentiometers – each to control the pitch of one of our buzzers.



Now, using analogRead(), store the position value from each potentiometer in a variable. You can see on the GitHub page that the named notes go from about 30Hz to 4900Hz (there is some more in-depth discussion of possible frequencies that you can read, as well).

map() your analogRead() values to the range 30 – 4900, so that the user can pick any of those available frequencies for either buzzer by turning the two potentiometers to different positions.